

Big Data & AI Transformation: **Shell Data Strategy**

| Trinity College Dublin
| Big Data & AI in Business
| MSc Business Analytics

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1. Business Objectives

Shell PLC is the second largest oil company in the world. Headquartered in London, it has operations in over 70 countries. As an oil company, Shell is vertically integrated through all aspects of the oil and gas industry, including: exploration, production, refining, logistics and transportation, distribution, power generation, and other petrochem products. Additionally, Shell runs JVs with other industry players and governments frequently, as oil projects are huge undertakings and cannot be operated by one player alone. To better explain the scale of operations:

- Shell produces 2.8 million barrels of oil equivalent a day and trades over 8 million
- Shell employs over 96,000 people and works with over a million business customers
- Shell sees 33 million customers at Shell branded retail stores every day, across 40,000 gas stations and 73,000 public electric vehicle charge points
- Shell has a corporate focus on sustainable, green and low-carbon technologies, and invests ~\$1.1 billion in that regard.

As a publicly traded company, Shell wants to maximise shareholder value. Additionally, it is bound by climate and environmental regulations of the countries it operates in and seeks to maximise its reach into sustainable technologies as governments across the world push for a greener future. This goal is encapsulated by Shell's core values:

- Powering Progress
- Respecting Nature
- Generating Shareholder Value
- Achieving Net-Zero Emissions

Shell is also intensely data-driven: out of its four major verticals - Upstream, Downstream, Integrated Gas and Renewables, and Projects and Technologies (P&T) - P&T focuses heavily on the technological aspects of daily and major operations, with offices across all major global tech hubs. P&T has over 12,000 people across global offices that work with data in some form to extract actionable ideas, plans, insights or projects that would add value to the company's operations. While these projects are varied, all use internal Shell or market data to achieve their goals.

While Shell does focus heavily on optimisation and streamlining all its processes in the name of efficiency, a company of this size will always have problems with scope creep, convolution, and overgrowth of projects and systems. These issues are what we will attempt to solve across two verticals: Upstream and Downstream. To support this, we have also created some new systems to handle human data.

2. Current Standing

2.1. IG & Upstream:

Shell Upstream is the engineering heavy side that deals with the exploration, extraction and processing of crude specifically, since natural gas is handled by IG&R. Shell operates in deep sea and near/on-shore drilling as an integrated company, but also works with third-parties in some regard. Usually, projects involve companies that deal with:

- Services: Exploration, Labour, Other services
- Constructors
- Equipment
- Drilling
- Security

Additionally, all projects are required to have environmental and legal clearances, which often involve multiple countries and take a lot of data and scrutinisation to realise. The process for getting an IGU project started takes years. M&As are also very common in this domain.

2.2. Downstream:

Downstream involves refining, logistics, marketing and distribution of the produce from Upstream. In this, Shell handles the production of gasoline, LNG, other fuel products, petrochemicals, lubricants etc. Supply chain management, retail and distribution and B2B and B2C marketing are also major parts of this domain.

Downstream is more dynamic, since the environment it operates in is rapidly changing. There are also many more variables at play compared to upstream. While more complicated, these variables are generally very well-defined in scope.

2.3. Human Data:

Shell employs over 90,000 people and serves over 33 million people daily. Shell generates massive amounts of data, internally and externally. While human data is collected and processed by HR and marketing, it is usually done with the objective of optimisation and profit, and is often unstructured. With these scales, the potential insights derivable from the data available are much greater than actually realised, beyond internal HR functions and advertising. An entrenched, complicated organisational structure also means that the insights or action items generated by said data can often be ignored by the people tasked with implementing them.

2.4. Current Architecture

As discussed, Shell collects huge amounts of data across many layers and verticals. One of the biggest challenges it therefore faces is redundancy and accessibility. Teams are often cross-functional and require data from across verticals, often from databases they might not have access to. Shell's organisational structure also holds a lot of sway over this as teams rarely have visibility of other teams and their work. If a cross-functional team requires extra-functionality data, there are many time-taking checks that slow down operational speed.

Shell uses SAP as its single-source-of-truth ERP software to track everything. The sheer scope and complexity of this database and associated SAP functions requires multiple teams just to maintain it. SAP is also notoriously difficult to use for people not trained in it, thus making access to the data dependent on focal choke points. Day-to-day data flows in and out of SAP systems are broadly automated, but can depend on focal points for manual data uploads, thus tightening slack times. Again, the scale of operations leads to data fabrics being fractured in terms of technologies used and compatibility, as well as the skills required to use them as teams' goals and evaluation metrics can often lead them to overspend or use non-compatible niche tools just to get an MVP.

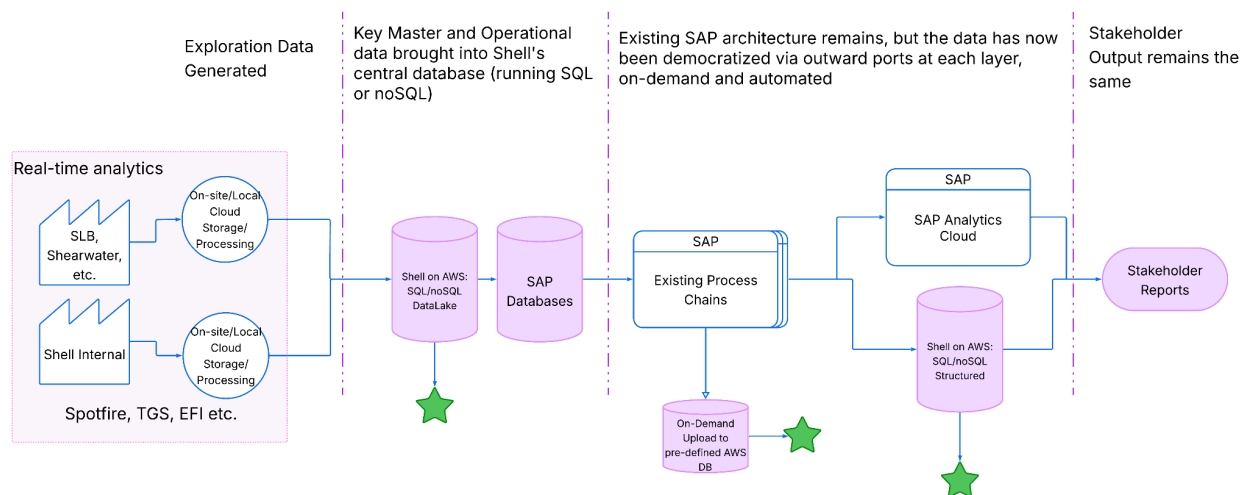
There are also issues with the placement of technological teams, primarily in India, as well as in its smaller controlling offices in London, the Netherlands and the U.S. Moreover, the operations Shell tracks and analyses are spread across the globe with unclear focals and contacts. The final data landscape is complicated and deeply disorganised. With SAP at the center, major arteries draw data from it, but splinter into disconnected nodes and capillaries, further complicated by employee dynamics and bad organisational structuring.

3. Upstream

Upstream is primarily a heavy engineering endeavor. As with any large-scale industrial project, there are many regulations and certifications governing its operations, even on a day-to-day basis. Moreover, there is a lot of data to be collected and crunched in early prospect exploration, lots of operational data to keep the wells running and a lot of higher-level data crunching to assess risk, value and progress for stakeholders. There are numerous considerations about the flow of data across international borders, local vs. global units of measurement, company structure and placement of employees, tools and technologies used: at which level and at what scale.

3.1. Data Architecture

Pre-Production:



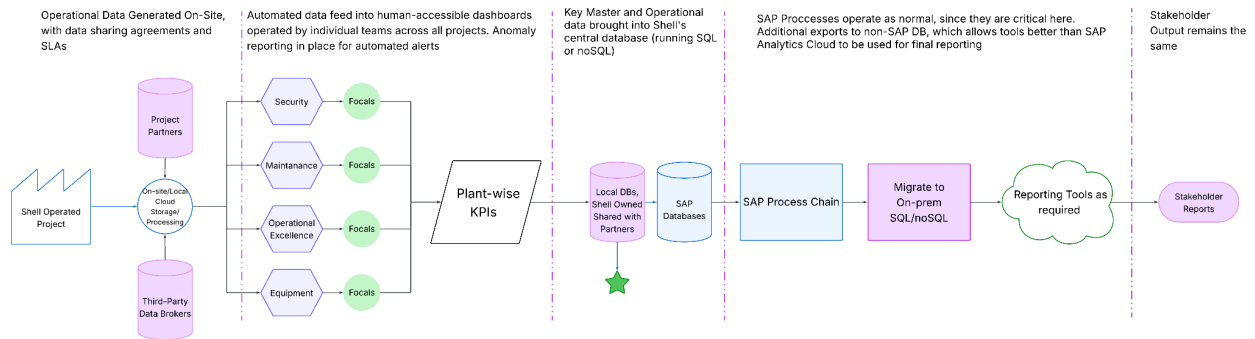
The pre-production architecture covers oil exploration and early-stage evaluation processes. Green stars denote access points for non-upstream teams. At this stage, data originates primarily from seismic surveys, geophysical studies and exploratory drilling activities conducted either by specialised third-party providers: seismic and subsurface service firms like Schlumberger and Shearwater, or by Shell's internal teams. These datasets are technically complex, high in volume and commercially sensitive, forming the foundation for reservoir evaluation and capital allocation decisions.

The first layer involves ingestion into a secure, Shell-controlled local cloud environment. This ensures controlled access, contractual compliance and protection of intellectual property before any wider distribution. Raw datasets undergo preprocessing steps like format harmonisation, metadata tagging, unit standardisation and quality validation. This layer is essential to ensure interoperability across geological, engineering and financial modelling teams given the diversity of contractors and tools involved. Then, structured data is securely transmitted to cloud infrastructure like AWS. This allows for a direct data draw-point for any other teams that might need it.

At key value-addition stages, offshoot databases are generated and managed by the data pipeline team. These allow broader organisational access without requiring direct interaction with core SAP tables. Analytical work at this stage is conducted using SAP Analytics, reducing export time to external tools. The final output supports executive decision-making, regulatory documentation and stakeholder reporting.

Active Production:

Operations



Active production has wells, drilling infrastructure and production facilities constantly churning out data. This includes pressure measurements, flow rates, equipment telemetry and other plant-level data points collected through operational systems and field sensors. This data is generated and owned across partner companies, and hence must be shared as such. Incoming data streams are validated and fed into anomaly detection programs locally due to the volume and operational nature of the data.

The data then moves into transformation layers where production data points are calculated and mapped to asset identifiers and site data. At this point, KPIs such as production rates, equipment efficiency and threshold values are derived and displayed to site managers and decision-making staff.

The production process is linked with maintenance data, cost accounting and compliance reporting in the ERP. The data-flow after is shared with the conventional SAP system as pre-production, with the exception of allowing batch-processed data to be used in third-party analytics tools (PowerBI) over SAP AC. The data then goes into reporting layers, which enable plant-level dashboards, operational analysis and executive performance reporting.

Strategic and Operational Benefits:

The new data architecture provides the following advantages:

- Tackles fragmentation through structured ingestion, standardisation and distribution of data, enhancing availability while preserving governance.
- Enhances security and compliance by ensuring that mission-critical data is processed in secure local environments before being integrated into worldwide cloud infrastructure and SAP systems.
- Increases decision velocity due to faster access to validated geological while feasibility information speeds up investment and approval cycles. In production,

real-time data validation and analytics enable faster operational decisions and reduce downtime.

- Finally, the new architecture reduces dependence on manual data extraction and technical limitations of using the SAP system. The structured offshoot databases enable technically adept users to analyse data effectively without compromising system integrity or increasing wait-time on focals.

4. Downstream

4.1. Shell's Downstream Supply Chain

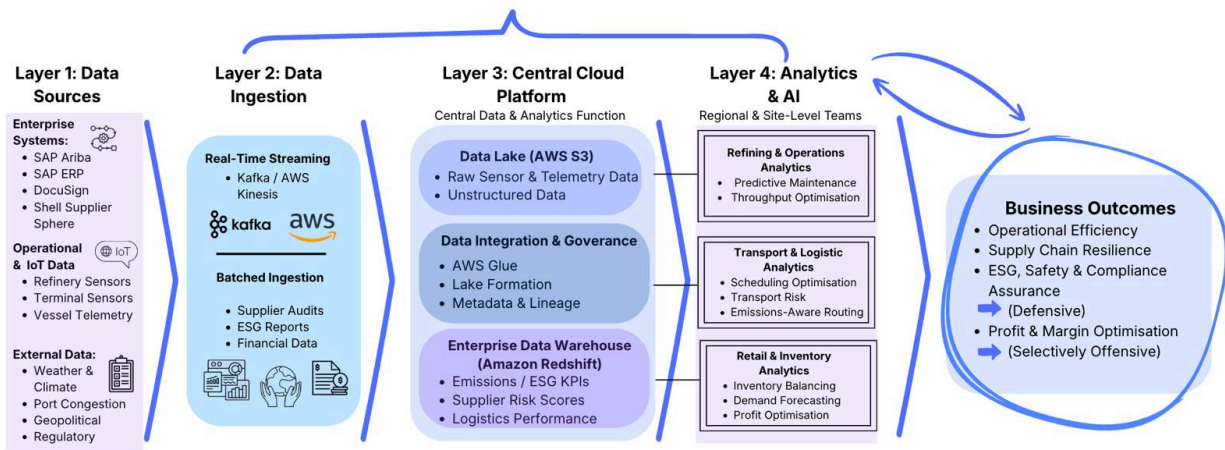
Shell's downstream business operation includes refining and distribution, and the channels that sustain it, including marketing. Shell's supply chain is complex, spanning numerous jurisdictions, and is centred around sustainability and CSR principles. The Shell Supplier Requirements prescribe a strong due diligence mechanism for:

- Business integrity
- Ethics and compliance
- Assessing Financial risk
- Health, Safety, security, environment and social performance
- Worker welfare, labour and human rights

Shell's Powering Progress Strategy also outlines its aim to transition towards net-zero emissions across suppliers.

Data Architecture:

The primary objective of this revised strategy is to treat data as a continuous flow rather than a static asset, enabling optimisation across a complex global system centred around three priorities: operational efficiency, supply chain resilience and measurable emissions reduction. This strategy is therefore primarily defensive, with selective offensive elements. It focuses on risk reduction, control, compliance and resilience throughout the supply chain. It focuses on risk reduction, control, compliance and resilience throughout the supply chain. However, it is also targeted towards maintaining a competitive edge through analytics, innovation and sustainability differentiation. As Shell increasingly embeds ESG, safety and ethics across its operations, the data strategy should be altered from retrospective reporting to real-time, data-informed control.



Data Sources and Ingestion:

Shell's downstream supply chain generates vast volumes of operational, transactional and logistical data across refining, storage, shipping and distribution. Shell already captures this via tools such as SAP, DocuSign and the Shell Supplier Sphere. Combined, these tools support an end-to-end lifecycle management system for any supplier relationships. Additional semi- and un-structured data from telemetry sources, partners' data and employee uploads, and can be complemented by external data sources such as third-party datasets on geopolitical risk factors, weather reports and industry-wide supply and demand data etc. from data analytics companies such as Wood Mackenzie.

Data ingestion prioritises capture and local analysis using real-time streaming technologies that ingest high-velocity sensor and shipping data, while batched pipelines handle supplier audits, ESG considerations and financial reporting.

Data Storage and Technology Architecture:

Country-specific regulations may govern local or cloud options, but a hybrid data architecture blending a centralised enterprise data lake with domain-specific databases or marts on AWS is particularly well-suited as a proven corporate strategy. AWS tools allow for storage, management and extraction as needed, while IAM creates a data management layer that will ensure data standardisation and quality, lineage, access control and ownership. As a principle, high-volume ops data is processed in resilient distributed environments, while batched data is sent to central repositories to be used for reporting dashboards.

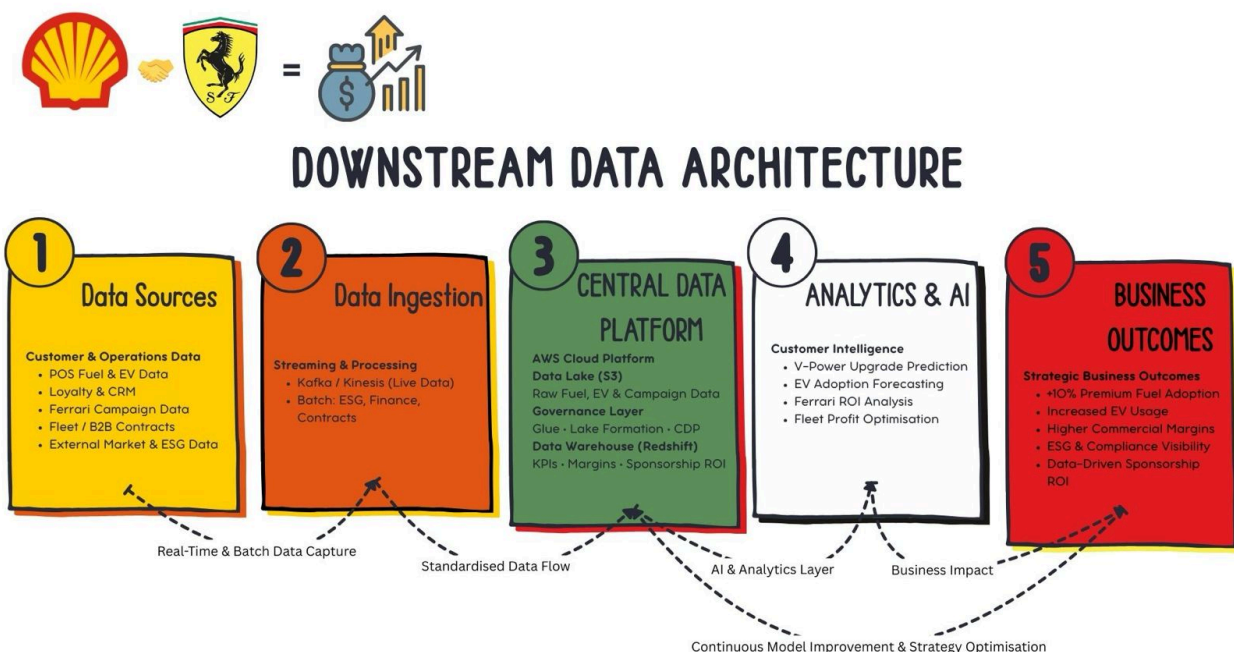
Central governance keeps metrics such as emission intensity or supplier risk scores consistent, while downstream teams can develop local analytics tailored to their contexts. Shell can now deploy predictive analytics, optimisation models and decision-support systems across supply chain activities. Crucially, outputs remain data-informed, rather than

fully data-driven. This ensures human oversight in key safety-critical and regulatory-sensitive decisions to support the Shell Supplier Requirements, while still benefiting from AI efficiency.

4.2. Leveraging the Ferrari Partnership

Shell's downstream strategy also aims to increase premium Shell V-Power fuel and EV charging adoption by 10% within the next 12 months. We can leverage its global partnership with Ferrari as a brand ambassador for performance, innovation and technological excellence as Shell V-Power fuels are co-developed in motorsport environments.

However, Shell currently lacks a fully integrated system to measure how Ferrari-driven engagement translates into downstream revenue growth. This strategy therefore focuses on converting brand equity into measurable commercial outcomes through customer intelligence analytics.



Shell already produces extensive amounts of downstream operational data. This includes:

- Point-of-sale systems at Shell Mobility Stations and EV Re-Charge platforms to track customer metadata.
- Integrated SAP systems which provide high levels of data traceability for reporting and audits.
- CRM and Marketing analytics tools that can crunch customer engagement and purchase patterns, at stores, or online.

- Datasets from external brokers to supplement customer data.

We can utilise the existing downstream data pipeline already established.

Data Storage and Technology Architecture:

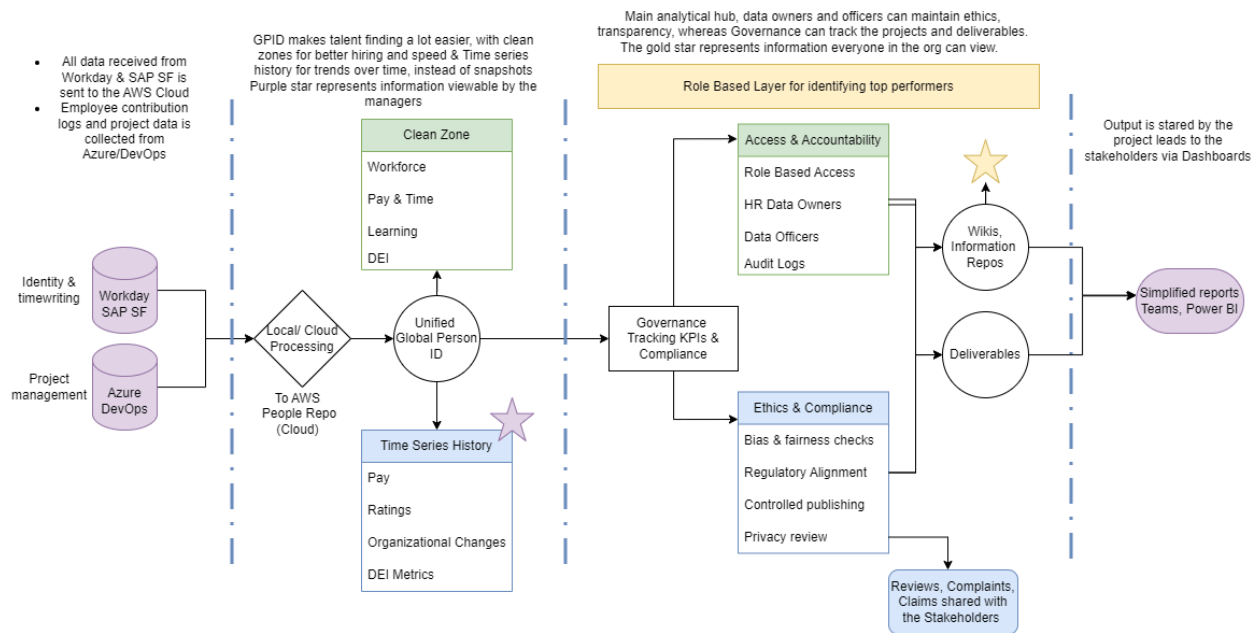
AWS continues with a retail-specific central enterprise data warehouse to standardise currencies, units of measurement and regulatory compliance across jurisdictions. AWS allows easy integration of other pipeline tools, such as:

- Real-time data streaming technologies, enabling live ingestion of fuel and EV transactions
- A Customer Data Platform, unifying identities across fuel, EV, retail and digital channels, creating a single customer view
- Advanced analytics and ML tools, enabling reporting and predictive modelling and segmentation

This identifies customers who will switch from regular fuel to V-Power after they see Ferrari advertisements using predictive analytics enabled forecasting and customer behaviour driven brand management and advertising.

This system runs in parallel to the SAP lines. Scalable, easy-to-use AWS tools allow secure but quick access for any cross-functional teams with little downtime and red tape. To complement this, a hybrid governance model is recommended. Centralised teams should manage global architecture, AI development and compliance oversight to ensure consistency and data privacy adherence (e.g. GDPR). Decentralised regional teams should execute local campaigns and pricing strategies based on market conditions. This will allow for Shell/Ferrari brand dispersion into every retail customer's buying decisions.

5. Human Data



5.1. Identity

Shell currently relies on Workday and SAP SuccessFactors for worker identity and timewriting. This dissociation makes redundancy and integrity an issue as an additional layer of complexity is introduced to managers and employees. The lack of a mandatory use policy regarding project management tools, such as Azure DevOps or Jira, for each team and project also results in employee contributions to go unnoticed, thus reducing transparency in calculating performance metrics.

Data is also siloed into verticals, with its storage across SAP, Workday and more usable Excel platforms making it harder for managers and HR decision-makers to stitch together. Hence, hiring, performance and development decisions can often be made on incomplete or inconsistent data.

5.2. Store & Integrate

Introducing a unified Global Person ID to track employees and contractors across projects and verticals, even outside the company, makes identity management, resource upskilling and finding talent a lot easier, while also reducing attrition as contracted and in-house employees are visible in success and risk analytics.

Backend databases can be integrated with HR, PM and SAP tools to clearly track performance. Making this data process transparent reduces manager bias and promotes Shell's DEI goals. SAP timewriting integrates this GPID, which allows the central AWS

cloud-based People Repository to automatically reflect time to managers, while allowing billing to be handled by SAP systems.

A central repository also introduces clean zones like workforce, pay, time, DEI and learning. It enables time-to-hire analytics for better hiring quality and speed, and tracks talent development ROI. Time-series history for pay, ratings, organisation changes and DEI metrics helps model attrition risks and culture trends over time, not just snapshots. Once established as the main analytical hub, it also ensures regulatory compliance with consistent, auditable reporting.

Additionally, data uploads made under a GPID through a custom-built interface integrated with all backend tools also establishes a role-based layer on top, thus allowing for person and role tracking and analytics to identify and reward top-performers, as well as motivate under-performers more accurately. This also frees up time for HR and managerial teams, allowing them to nurture and support employees dealing with demanding personal challenges, while also helping them to perform well.

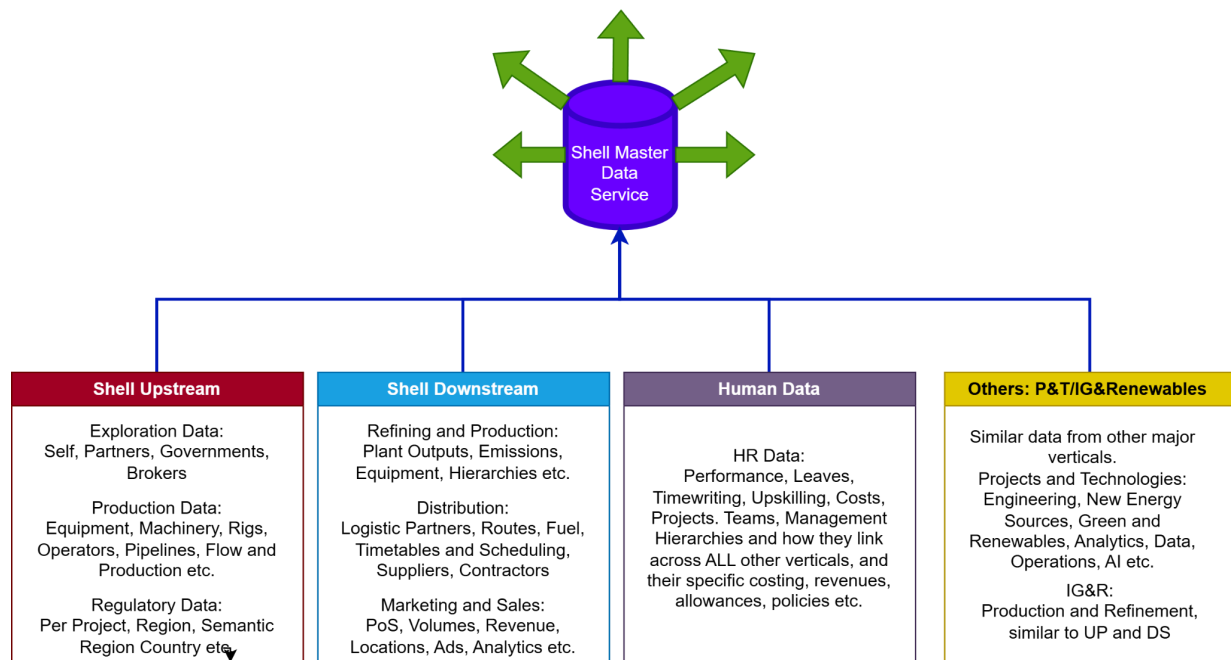
5.3. Governance

Managerial performance, reviews, complaints and claims can also be transparently tracked or obfuscated as needed through simplified reporting dashboards built on PowerBI and Microsoft Teams tools. Governance can also be simplified with clear role-based ownership of projects, deliverables, people and data. HR Data Owners and Officers would maintain ethics, transparency and integrity of data generated by their employees.

Additionally, wikis and information repositories maintained by these roles would make this information clear and accessible to all employees too. In parallel, a privacy- and ethics-by-design review for HR analytics would ensure fairness, transparency and compliance, strengthening trust while improving hiring, workforce planning and development decisions.

6. Conclusive Strategy

The discussed business specific strategies are underpinned by a central Master Data Repository, built on SQL with a streamlined secure access route. This master data feeds in from every major business and SAP system, and makes its contents available to any team's business case.



This makes data generated and stored by Shell democratized and easy-to-use for all teams and employees, reducing lead and down-times in obtaining access, learning tools to extract or use data.

We also maintain standardised tools for operations: SAP acts as the single-source-of-truth used by everyone, but the data uploaded there is tagged appropriately to make it easily trackable by anyone. There are well-defined data outlet points at each value-add level to ensure access to non-SAP-proficient teams via (no)SQL. Analytics and visualisation tools can be used as required, but output formats and datatypes remain consistent.

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